



ULTRA THERMO SCIENTIFIC

Complete Environmental Simulation Systems

CLIMATIC TEST CHAMBER OPERATION AND MAINTENANCE MANUAL

MODEL NO.: UT/CTC 1000-65-4

SERIAL NO.: UTS 22/22-23/9-22

ULTRA THERMO SCIENTIFIC

No.3A, Manimegalai Street, East Tambaram, Chennai – 600 059.

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CAUTION: Do not operate the chamber beyond the specifications.

CAUTION: To protect from temperatures that are too high or too low, you must use a protection device. Product protection is your responsibility.

1. PRODUCT SPECIFICATIONS AND OVERVIEW

ULTRA THERMO SCIENTIFIC Climatic Test Chamber Model UT/CTC 1000-65-4 is capable of conditioning 100 Liters of Volume space, to the programmed temperature values, from -65°C to +180°C, with highly reliable accuracy and control. The conditioned volume space can be used to expose the sample, to test its functioning at different temperatures.

Dimensions : Internal Dimensions : 1000 (L) x 1000 (B) x 1000 (H),
Volume : 1000Ltrs

Temperature Range : -65°C to +180°C,
Control Accuracy : within +/- 0.1°C to 0.8°C after stabilization
Temperature Gradient Uniformity : within +/- 1.2°C after stabilization
Cooling Ramp Rate : 2degC/min average between -45degC and +180degC
Heating Ramp Rate : 5degC/min average between -45degC and +180degC

Humidity Range : 10% to 98% RH +/- 2% RH at 20°C to 90°C, limited to 4°C dew point
(Dry Compressed Air at 3-4 bar pressure will be required for Dehumidification, at the site of installation).

Refrigeration Unit : Cascade Bitzer Germany, Air Cooled Condensing Unit complete
CFC Free refrigerant R404A/ R23.

Finish : 16SWG SS304 Quality Stainless Steel Fully TIG Welded Interior for
Vapor Tight Finish used in the Interior and 16SWG MS Powder Coated
In the Exterior of the chamber.

Tray : 2 No. SS304 Fabricated Loading Tray (80Kg Load Bearing).

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PRODUCT SPECIFICATIONS AND OVERVIEW (Contd.)

- Insulation** : 96Kg/m³ density Non-flammable Rockwool 100 mm thick between the inner and outer box and Special Ceramic Felt material will be used for Door beading.
- Port Hole** : 2 Access Port Hole, Ø75 mm with Silicone Closure on side of the wall.
- Fan System** : 2 No's. (Ø300mm) High performance externally mounted, Flange Mounted Motor will drive the internal fans through Extended stainless Steel shaft.
- Humidifier, Dehumidifier System** : Adequate Air Atomizers for equalized and consistent Humidification of the entire chamber volume, Warm water vapor generator with heating system, over-heat safe system Vapor storage Tank complete With blowing system will Be used in the chamber.
- Rotronics Direct measurement Platinum capacitive type sensor shall Use for the measurement and control of Humidity. Automatic DM Water Refill in the vapor generator if Connected to the DM Water Line.
- Controller system** : The Eurotherm 2604 high performance 2 Loop profile programming PID Controller with PC Interface and software shall be used. Over and under temperature Indicator shall be of standard make.

Main Features of the controller shall be

- RS232C PC interface - with remote logging inclusive of acquisition and configuration software.
- High Stability Control & High Precision input.
- Gain scheduling, Auto tune.
- High accuracy, high stability temperature and process controller.
- Dual 7-segment display (5 digit) of the process value and set point with an LCD panel for display of alarm messages, programmer and loop status information. User defined messages in the LCD panel simplify operation.
- Advanced control algorithm gives stable straight-line control. Auto tune simplifies the commissioning procedure.
- Incorporates a self-correcting input circuit (INSTANT ACCURACY) to preserve & maximize the accuracy, performance during warm Up and change in ambient temperature.

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PRODUCT SPECIFICATIONS AND OVERVIEW (Contd.)

Door Assembly	: Door System with full front opening type, compensating hinges with Locking mechanism. Incorporated in the center is the 400 x 400mm 5 Pane viewing window, with the whole being insulated with non-flammable Rockwool and with an exterior finish, matching the main Cladding. Viewing window will be provided with sample viewing light. Strip heater for Viewing Window and Door Assembly will be provided. Inter look opening of door will be provided.
Safety features	: ➤ Door Interlock ➤ Emergency Switch ➤ Input supply variation >10% ➤ Over temperature protection ➤ Circuit breaker ➤ Thermal overload relay for Fans and Compressor
Other features	: ➤ Drain for condensate and cleaning water will be provided. ➤ External Storage tank for DM Water will be provided. ➤ Earth point will be provided. ➤ The power consumption by safety & control devices will be as low as possible. ➤ Energy efficient motor will be used (Siemens/Alstom). ➤ All the electrical items used shall be of std. IP 55grade suitable for Humidity chamber (Schneider Electric). ➤ All the switchgears and MCB's used in the construction shall be of high quality reputed brands (Schneider Electric). ➤ Noise level will be with 75 dB (A) at 1Mtr distance in front of the chamber, from the door.
Calibration	: Factory Calibration certificate will be issued which has Traceability To NABL for the complete system.
Display	: The programmer will show measured variable value in °C, and Humidity in %RH.
Electrical Supply	: 415 V $\pm$ 10%, 3 Phase, 50 Hz AC + Neutral + Earth 40A 4 Pole MCB will be required at the site
Working Ambient	: Peak Specifications confines to the working ambient of 45°C, 90% RH.

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Environmental Conditioning Functions:

Heating of the chamber is achieved by recirculation chamber air through open-air Nichrome wire heater elements. The elements are supported by ceramic insulators.

Air circulating is generated by one or more conditioner motors with either propeller type fans or blower wheels. Conditioner motors are externally mounted.

2. HEATER UNIT

Heating of the chamber is accomplished with the use of open air low mass nichrome wire heating elements. These elements have low thermal lag and provide rapid response to the controller's demands. The elements are mounted in the plenum conditioning section. This isolates them from the workspace and prevents direct radiation to the product. Electric power ratings for your chamber's heater bank are listed on the Electrical or Power Schematic.

Specification of heating coil

Dry Heater

Maximum Temperature	:	800°C
Sheath	:	Ø 7.9mm
Cold end	:	50mm
Termination	:	4 ba thread
Rated Voltage	:	240 VAC
Watts	:	1.5KW
No. of heaters	:	6 Nos.

RH Heater

Maximum Temperature	:	800°C
Sheath	:	Ø 7.9mm.
Cold end	:	50mm
Termination	:	4 ba thread
Rated Voltage	:	240 VAC
Watts	:	1.5 KW
No. of heaters	:	2Nos.

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Air-cooled BITZER Refrigeration compressor is used in this chamber

REFRIGERATION SYSTEM (Low Stage)

COMPRESSOR : BITZER COMPRESSOR
Model No: 2CES - 3Y - 40S
Serial No: 1602607732

CONDENSER : Air Cooled

DRIER : DML-053S

EVAPORATOR : Finned Type Evaporator

REFRIGERANT : R-23a

REFRIGERATION SYSTEM (High Stage)

COMPRESSOR : BITZER COMPRESSOR
Model No: 2CES - 3Y - 40S
Serial No: 1602607785

CONDENSER : Air Cooled

DRIER : DML-165S

EVAPORATOR : Finned Type Evaporator

REFRIGERANT : R-404a

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Humidifier and De Humidification UNIT

Humidifier system :	Air Atomizers for Humidification, Warm water vapor generator, complete with blowing system, heating and over-heat safety system.
Humidity Measurement :	Rotronics Direct measurement Platinum Resistance type sensors shall used for the measurement and control of the Humidity.
De Humidification :	Compressed Dry air at 3 to 4 bar pressure for De-humidification is required to be provided at the site of installation.

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3.0 OPERATION PROCEDURE

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Sequence of operation

This manual is provided to acquaint operation of the chamber. The chamber is completely self-contained and inter-connected, capable of producing and maintaining the environmental conditions.

Front Panel Control

Mains	Glow light when 3 Φ Supply on R Y B
Phase Failure	Turn off the chamber if any one of the phase in the mains.
Blower	Turnoff the chamber in case of overload of chamber fan.
Dry Heater	Turn off the Heater Circuit
RH Heater	Turn off the RH Heater Circuit
Low Water	Turnoff the humidity set-up, when the water decrease the actual level.
H/S Hp.	Turnoff the chamber in case of discharge pressure of Compressors increased above the specified limit.
H/S Lp.	Turnoff the chamber in case of suction pressure of compressor decreased below the specified limit.
L/S Hp	Turnoff the chamber in case of discharge pressure of Compressors increased above the specified limit.
L/S Lp.	Turnoff the chamber in case of suction pressure of compressor decreased below the specified limit.
Heat	Glow in Heat Cycle
Cool	Glow in Cold Cycle
Humidity	Glow in Humidity Cycle
De-Humidity	Glow in De-Humidity Cycle

Switches

Mains	Used to Start the overall chamber
Control ON	Used to start the Heat / Cold Cycle
Humidity ON	Used to start the humidity Cycle
Emergency Stop	Shuts down the chamber instantaneously
Push Button	Switch used to turn ON the Chamber light power supply for illumination.
RS485	System Interface

Display

Controller	Used to Program the temperature / Set & Process value
Hour Meter	Used to verify total running hours of chamber.

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4.1 EUROTHERM - 2604 CONTROLLER PROGRAMMER SETTINGS

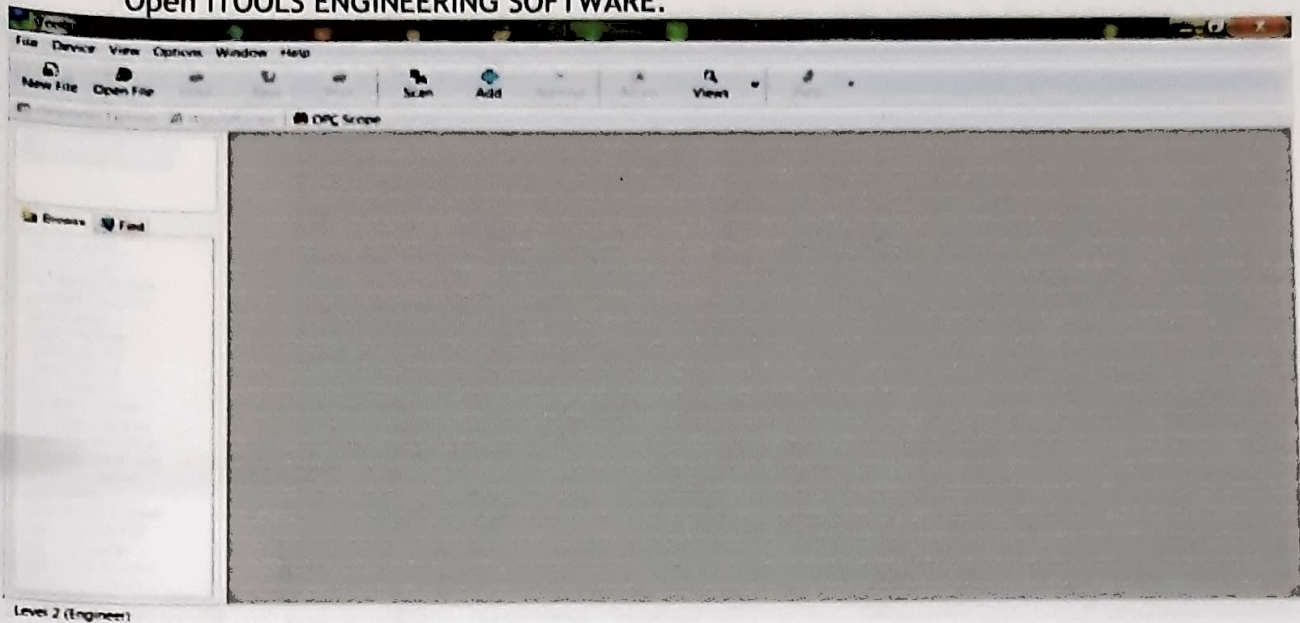
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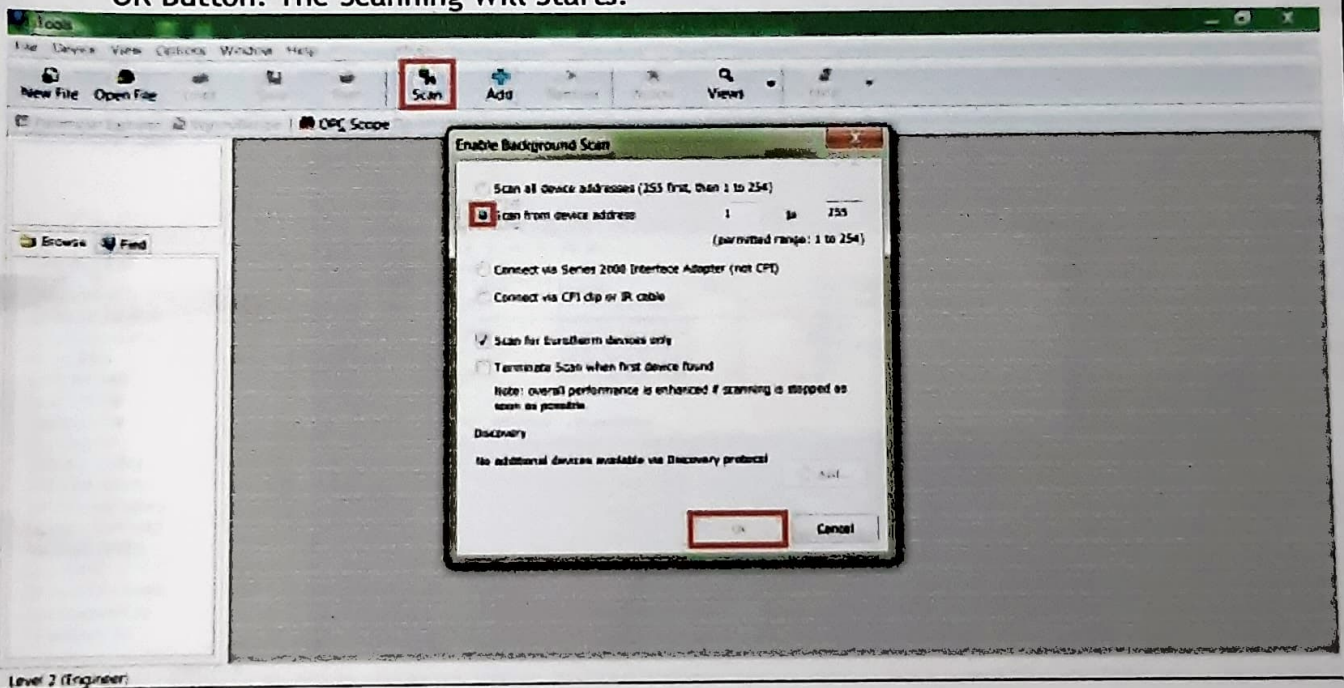
STEP 1:

Open ITOOLS ENGINEERING SOFTWARE.



STEP 2:

Then Click the SCAN option. It Opens the Enable Background Scan Menu Now Click OK Button. The Scanning Will Starts.



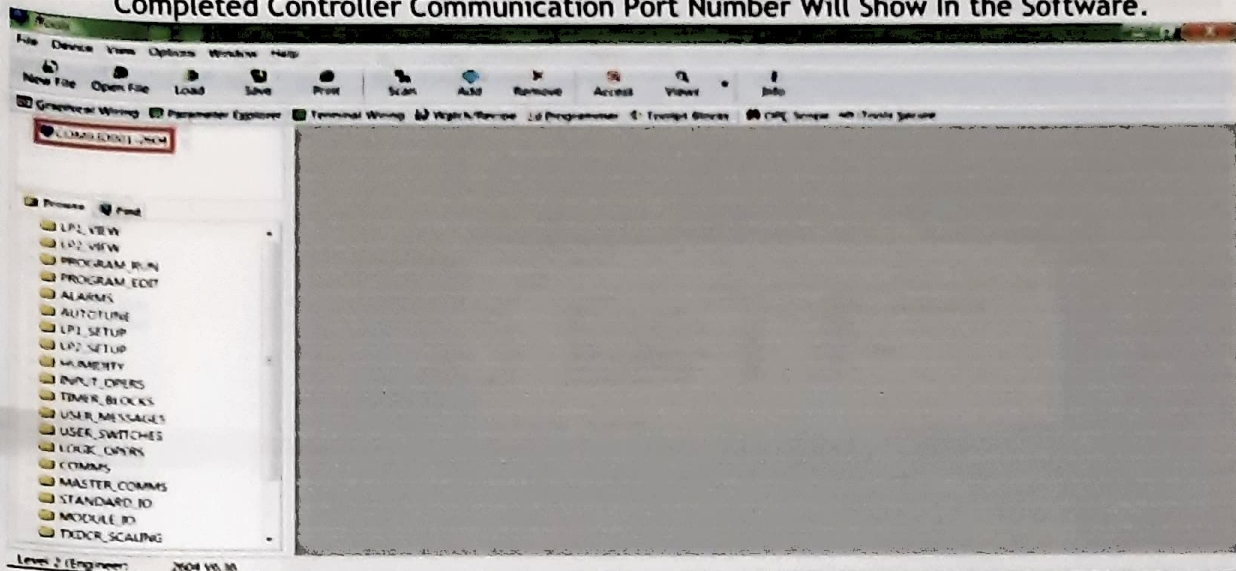
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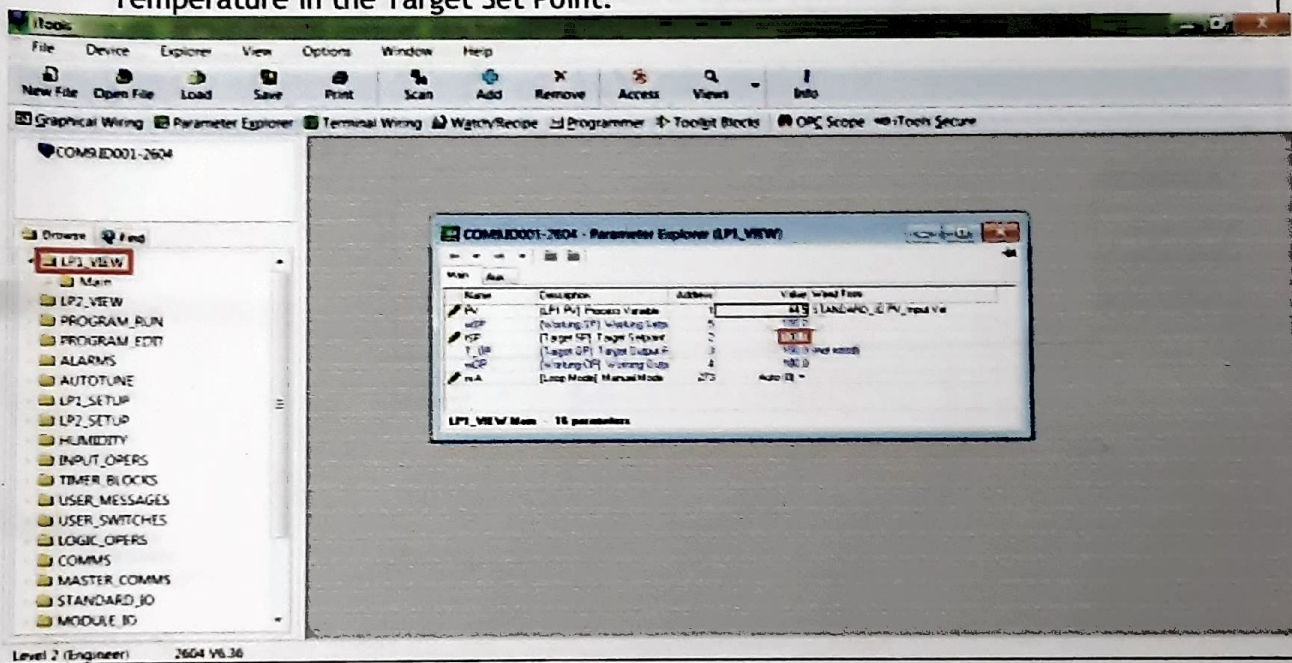
STEP 3:

After scanning, it takes Few Minutes to Synchronizing. After synchronizing Completed Controller Communication Port Number Will Show In the Software.



STEP 4:

Double Click the LP1_VIEW it opens The Parameter Explorer You Can Set The Temperature in the Target Set Point.



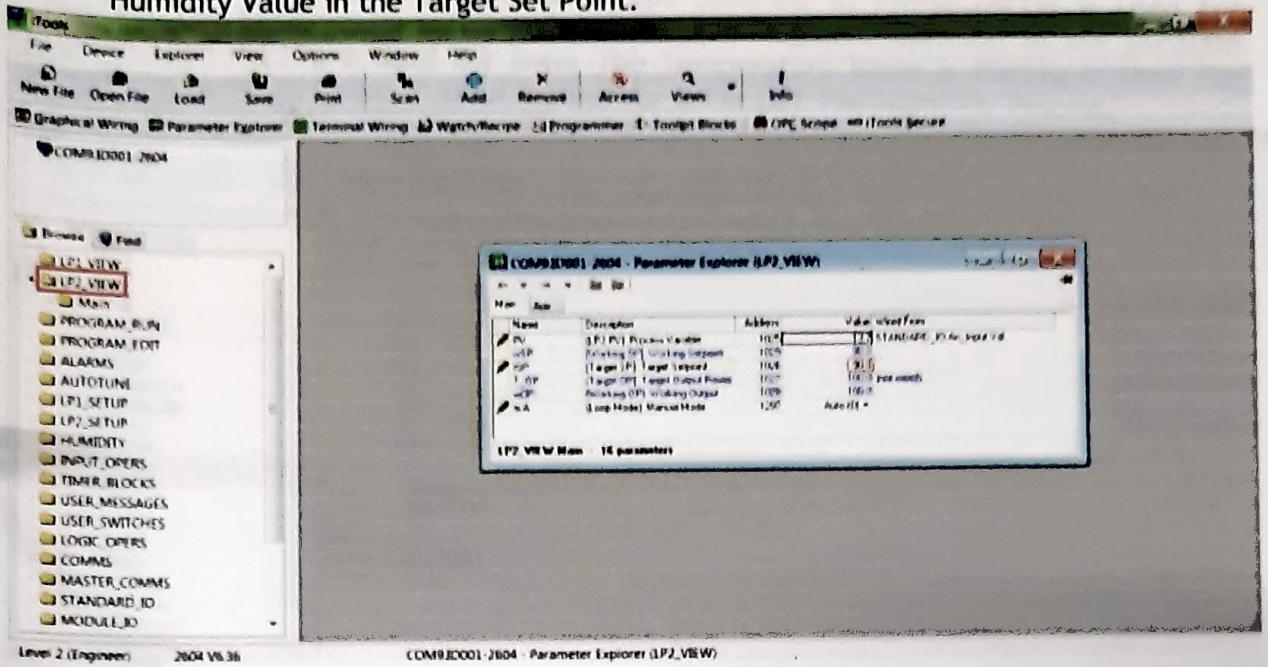
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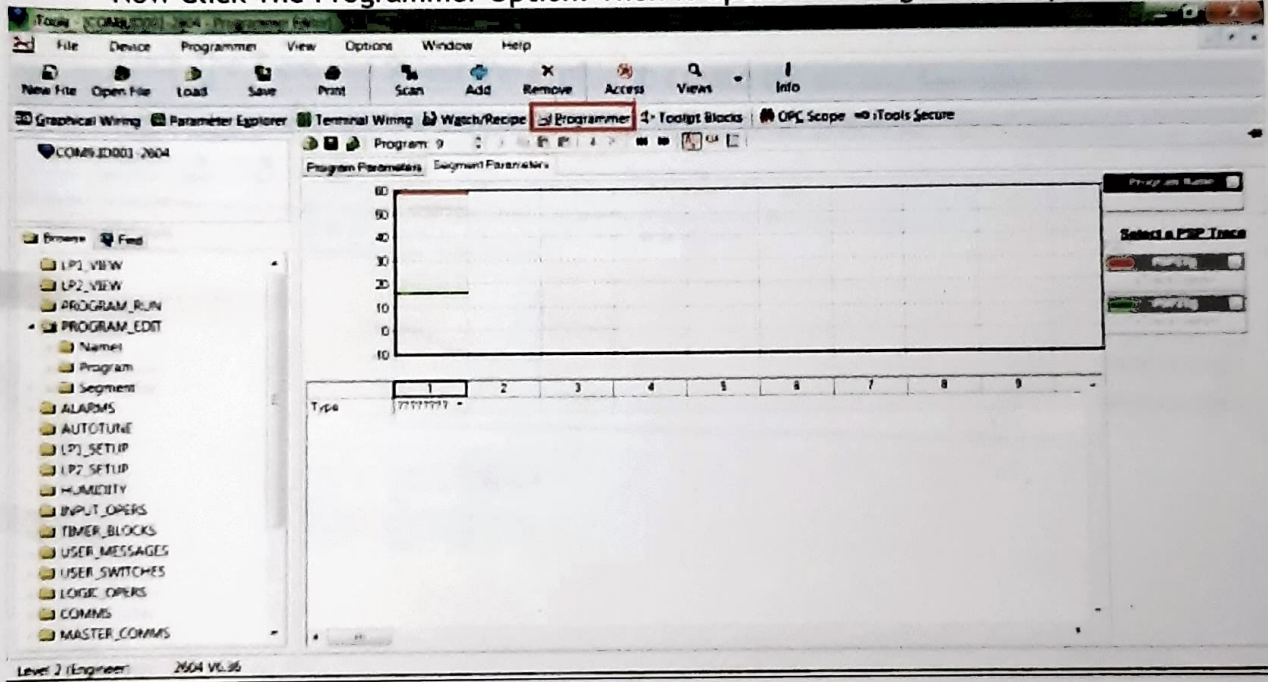
STEP 5:

Double Click the LP2_VIEW it opens The Parameter Explorer You Can Set the Humidity Value in the Target Set Point.



STEP 6:

Now Click The Programmer Option. Then It Opens The Programmer Option.

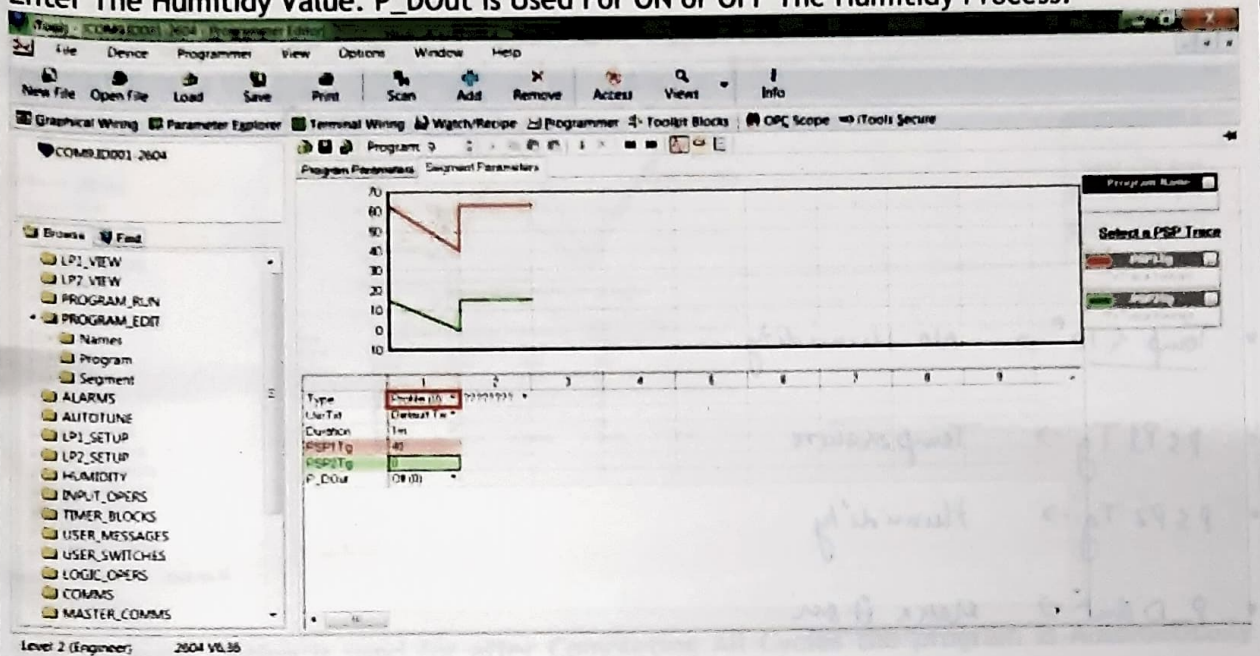


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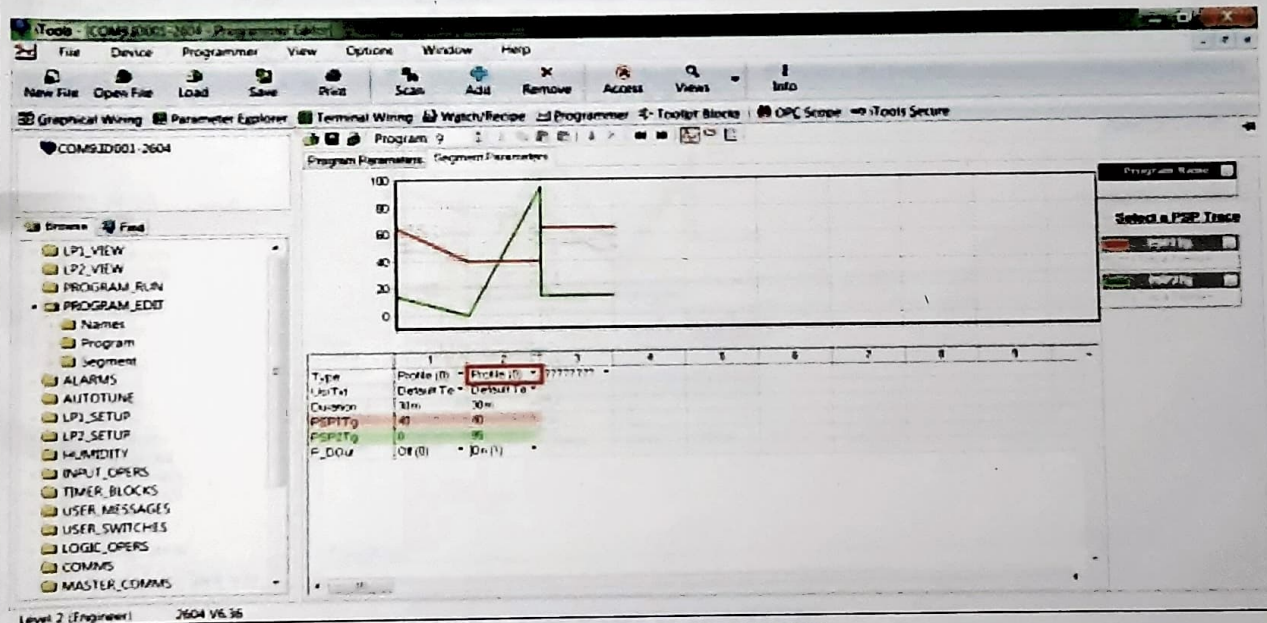
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In The Programmer Option Segment 1 Type Selected to Profile.UsrTxt is Used For Mention The Segment Name. You Can Enter the Segment Duration In the Duration Option.Then PSP1Tg is Used For Enter The Temperature Value & PSP2Tg is Used For Enter The Humidity Value. P_DOut is Used For ON or OFF The Humidity Process.



Click The Type and Select the Profile to create the second Segment.



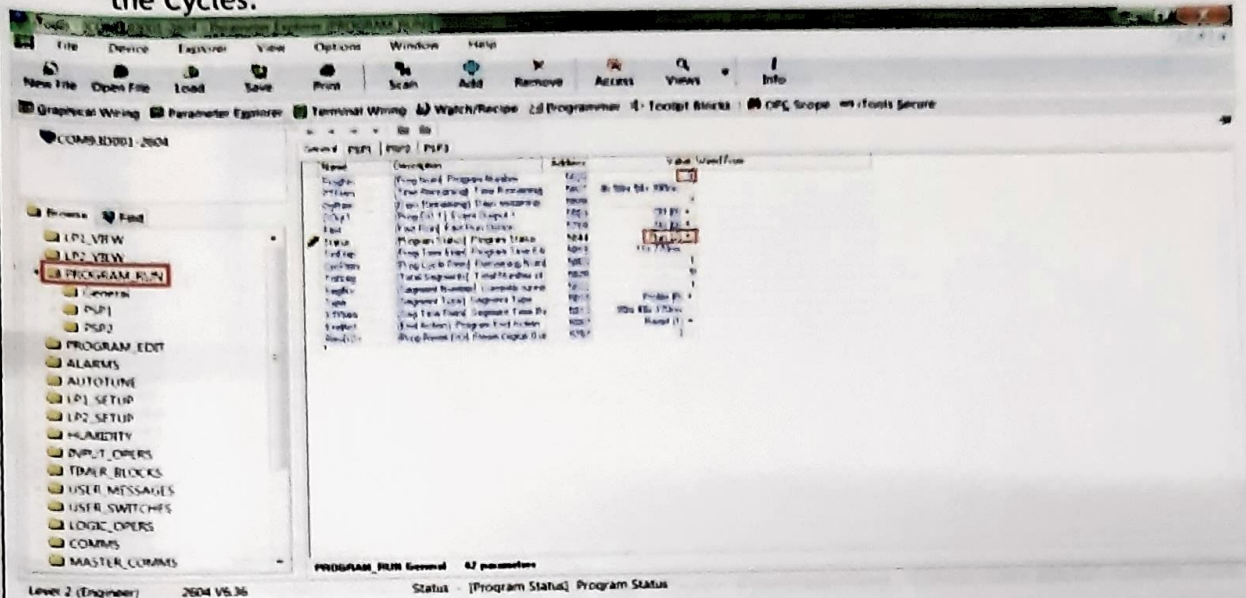
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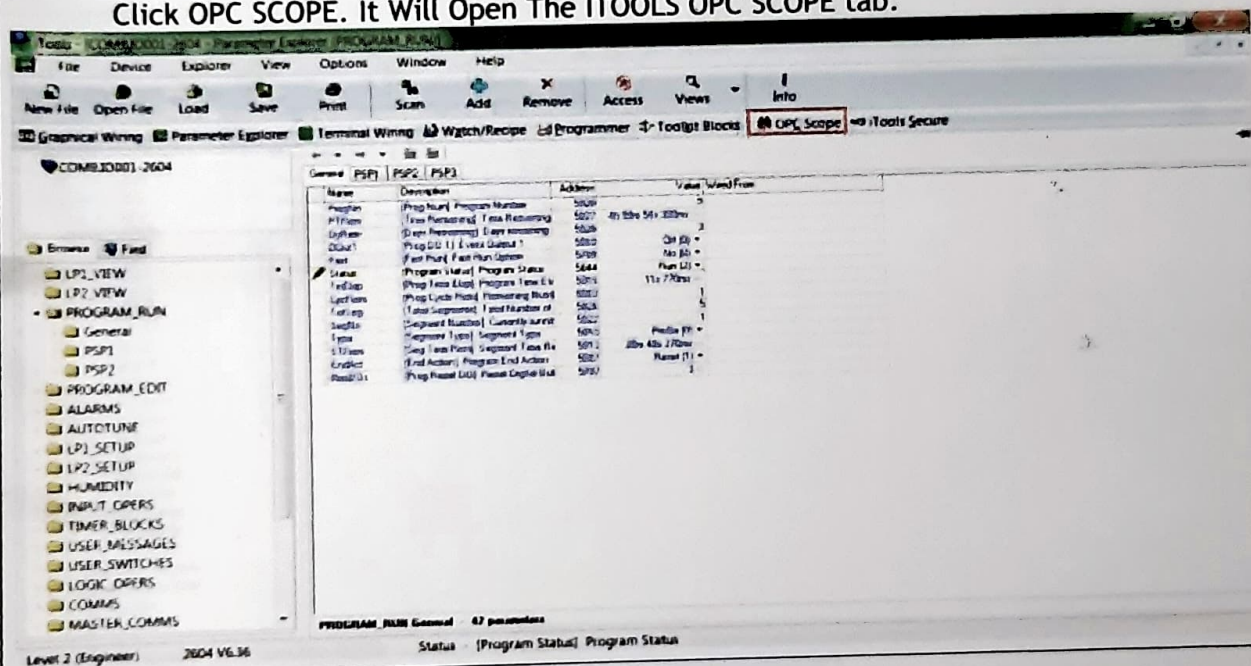
STEP 7:

Now Double Click the PROGRAM_RUN Option Enter the Program Number Same as the programmer Number Then Select the Run [2] Option in the Program Status. Now The Program is Starts. It Will Show The Time And Day Duration For the Cycles.



STEP 8:

Click OPC SCOPE. It Will Open The ITOOLS OPC SCOPE tab.



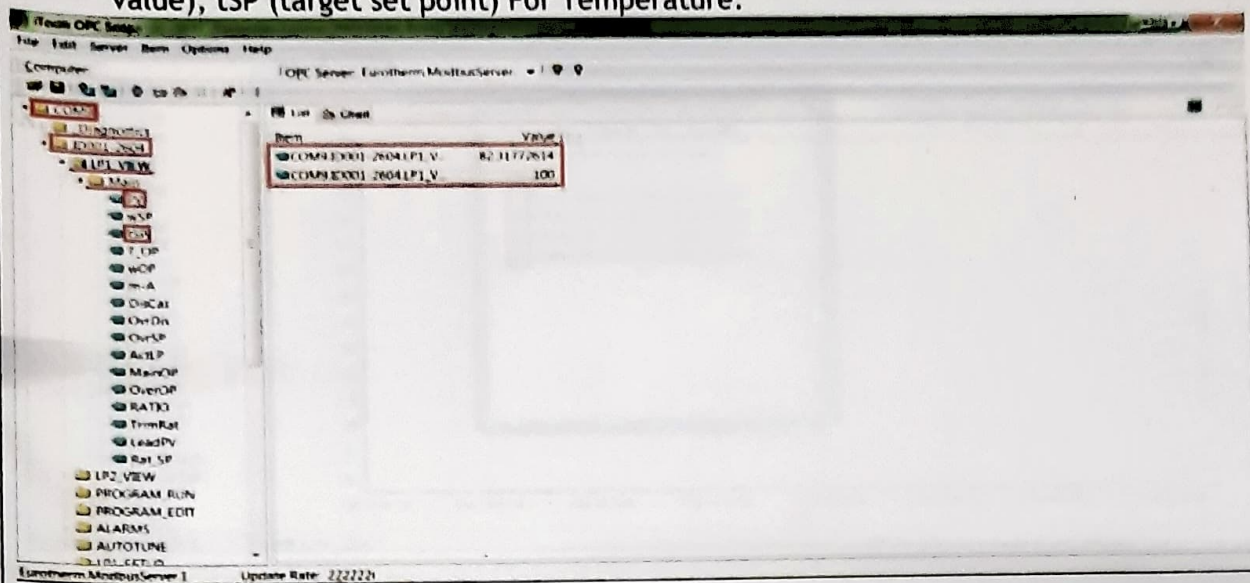
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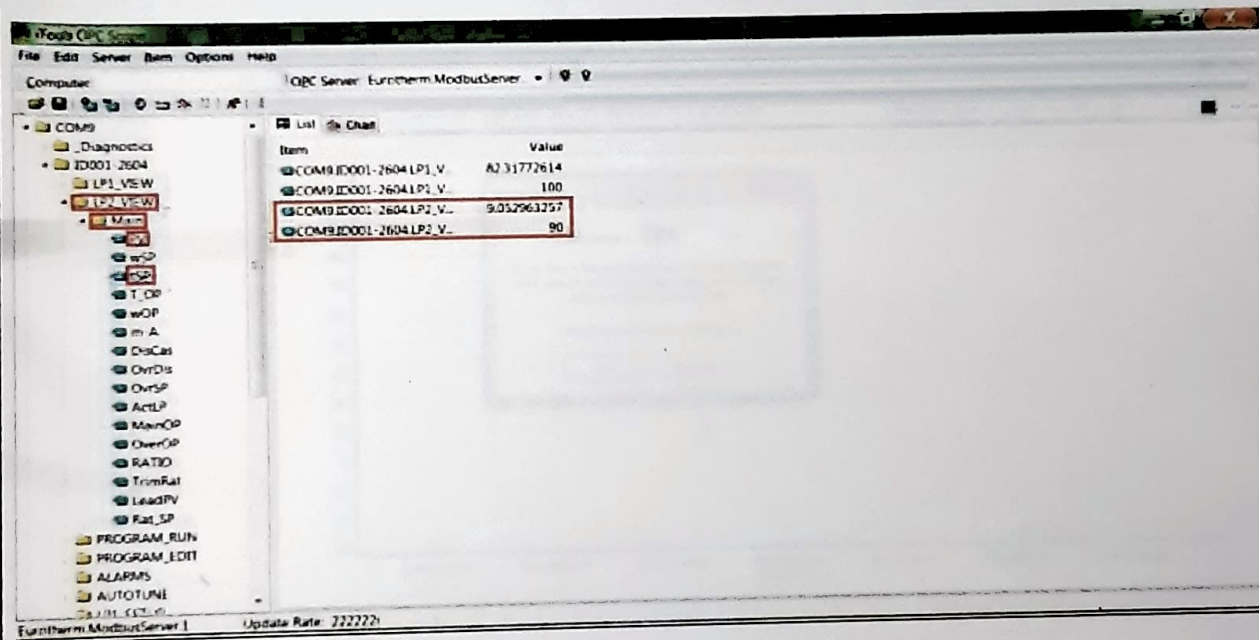
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STEP 9:

Select the Communication Port as Same as the ITools Engineering. Click the ID001- 2604 Then Click the LP1_VIEW Then Main and Select the PV (present value), tSP (target set point) For Temperature.



Select the Communication Port as Same as the ITools Engineering. Click the ID001-2604 Then Click the LP2_VIEW Then Main and Select the PV (present value), tSP (target set point) For Humidity.

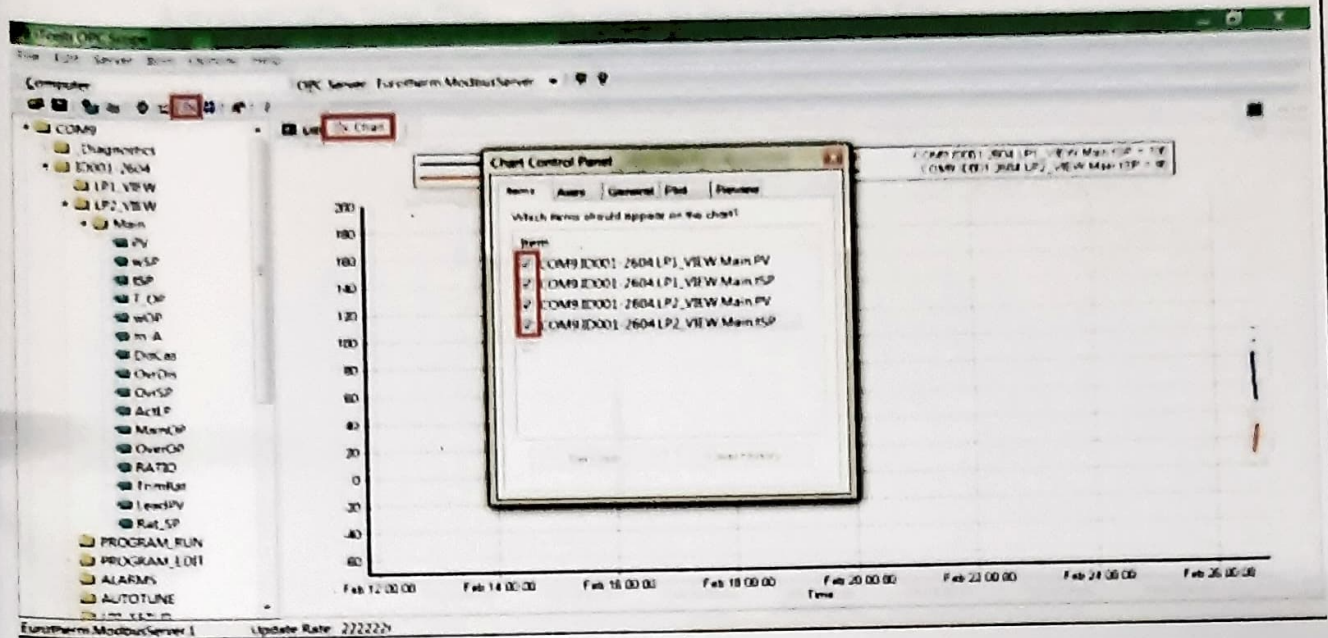


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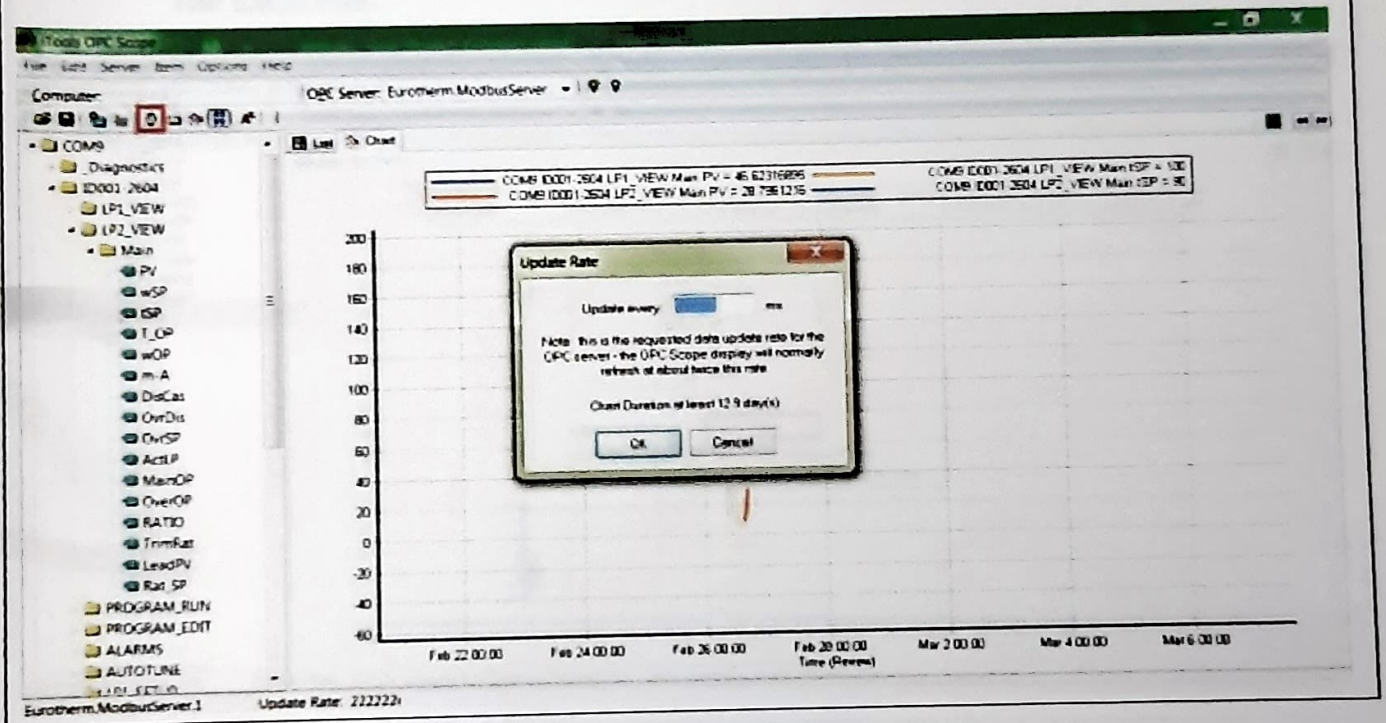
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Now Click The Chart then Chart Control Panel Select Above Items.



Update Rate Option is used for Change the Chart Duration Update Time.



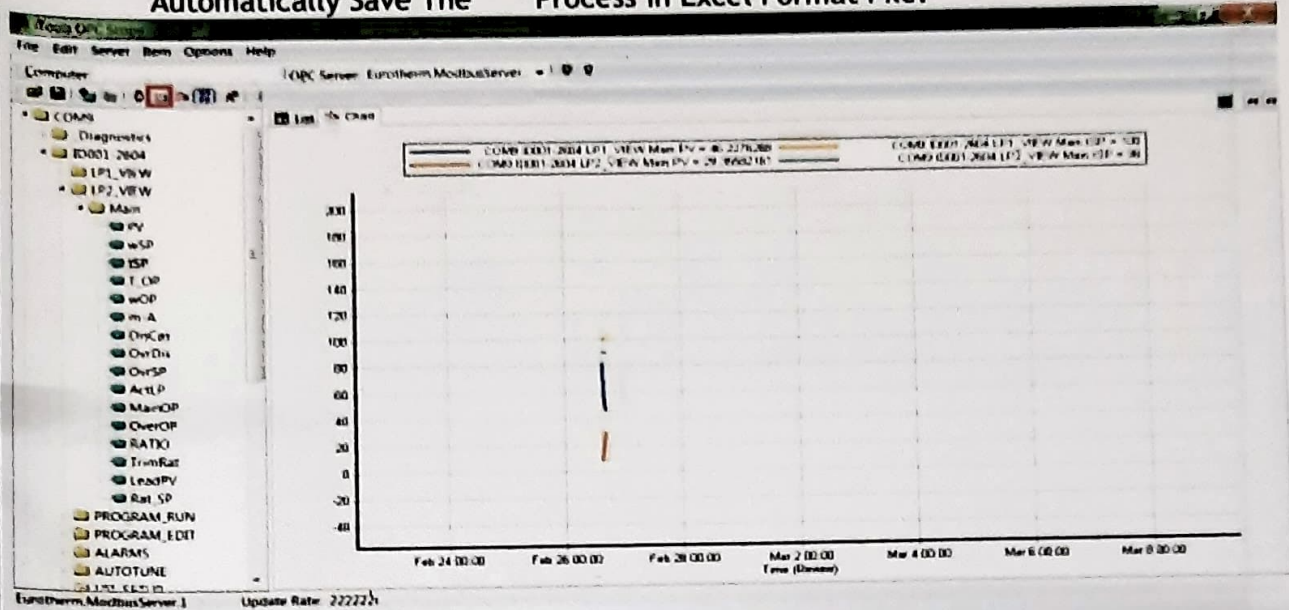
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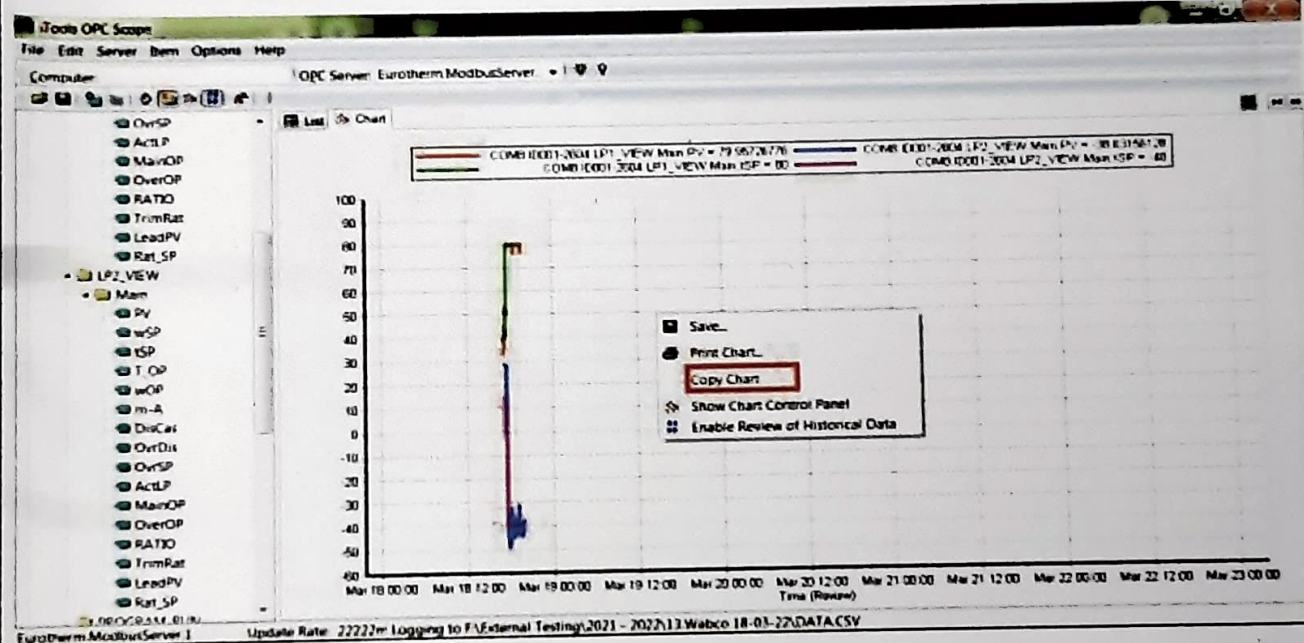
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STEP 10:

Click the Data Logger Select the File Location and Give File Name It Will Automatically Save The Process in Excel Format File.

**STEP 11:**

To Save The Chart Right Click On The Chart And Click Copy Chart And Paste in The Excel File.



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